# Experimental Evaluation of Robust Feature Selection

12 October 2026

**Abstract**

We study whether choosing a few variables by minimizing a robust testing loss preserves the difference between two populations. Four synthetic mechanisms distinguish testing power from signal recovery; independent prediction and two real datasets provide further checks. The recorded robust-risk optimum does not consistently give the best testing power or prediction. On reused small-case settings, retaining fraction bounds changes full subset evaluations by -25.8% and total median-summed time by +18.1%. This is an ablation, not independent confirmation.

## 1 Experimental question and common protocol

Suppose that we may observe only three of twelve variables. Which three should we choose so that two populations remain distinguishable? We minimize the worst-case testing loss over two empirical Wasserstein balls. We evaluate detection of distribution differences, recovery of the responsible variables, prediction on new observations, and the cost of subset search.

The statistical experiment contains three informative variables and nine noise variables. The first population has independent standard-normal coordinates. Only the three informative coordinates change in the second population, as shown in Table 1. Their positions are shuffled before selection. Each method chooses three variables from 20 observations per class. Selection, subsequent classifier fitting, and final testing use disjoint samples; a method cannot inspect the final observations when choosing its variables. For each type of difference, the complete selection and evaluation are repeated 30 times with newly generated samples. Figure points show the average result; intervals are 95% seed-bootstrap intervals. Parameters are collected in Section 7.

**Table 1:** Four ways in which the informative variables differ. All other variables have identical distributions in both classes.

| Difference | Change in the second population |
|---|---|
| Mean | Each informative coordinate has mean 0.9 instead of zero. |
| Variance | Each has standard deviation 1.8 instead of one. |
| Dependence | Their pairwise correlation is 0.65; each individual marginal distribution is unchanged. |
| Mixture | A shared random sign times 1.5 is added to all three coordinates; the means remain zero. |

Following Wang et al. [1], we evaluate testing, variable recovery, and computation separately. Our data models and fixed sample size and budget differ from theirs; their statistical parameter sweeps are not reproduced. The added dependence case has no individual-coordinate population difference.

### 1.1 What the methods select

Six practical methods receive the same three-variable budget. The global and greedy methods optimize the same robust loss; the other methods select variables using distribution differences, predictive coefficients, or random choice.

**Table 2:** Practical selectors. All methods select three variables; implementation parameters are collected in Section 7.

| Selector | How the variables are chosen |
|---|---|
| Global risk | Search over subsets, using lower bounds to rule out unpromising subsets. |
| Robust greedy | Add one variable at a time, refitting each candidate and choosing the smallest returned feasible robust risk. |
| MMD screening | Rank individual coordinates by a kernel measure of the difference between their two distributions. |
| Sparse additive MMD | Jointly choose sparse coordinate weights using the linear additive-kernel criterion of Wang et al. |
| L1 logistic | Rank the coefficient magnitudes of a logistic model with an absolute-value penalty. |
| Random | Draw a three-variable subset uniformly. |

Two additional references help interpret the results. *Known signal* uses the three true informative variables, which are unavailable to a practical method. *All features* uses all twelve variables and therefore exceeds the budget. These references show what happens when the signal is known and when it is kept together with all the noise. They are not budget-matched competitors.

Only Wang et al.'s linear additive criterion is implemented, with fixed parameters; this comparison is not a reproduction of their full experimental study.

The statistical results below describe subsets produced by the earlier search implementation. Section 6 evaluates the revised search separately on small optimization cases. Those new timings do not establish new statistical performance or large-scale speed improvements.

## 2 Can the selected variables detect a distribution difference?

We apply the same test to every selected subset. A kernel measures similarity: the Gaussian radial-basis-function (RBF) kernel $k(x, y) = \exp[-\|x - y\|_2^2/(2b^2)]$ is large for nearby observations; $b$ is the bandwidth. For samples $x_1, \dots, x_m$ and $y_1, \dots, y_m$ from the two populations, squared maximum mean discrepancy (MMD) compares three averages:

$$T = \operatorname*{avg}_{i \neq j} k(x_i, x_j) + \operatorname*{avg}_{i \neq j} k(y_i, y_j) - 2\operatorname*{avg}_{i,j} k(x_i, y_j).$$

The first two average within-class pairs, excluding self-pairs; the last averages all cross-class pairs. Larger $T$ indicates a greater distribution difference.

*Power* is the fraction of new batches on which a genuine difference is detected; larger is better. *False-alarm rate* is the rejection fraction when both batches come from the same population; it should not systematically exceed the nominal upper level 0.05; lower conservative rates are acceptable. We use an unbiased squared-MMD statistic $T$ and a permutation test:

$$p = \frac{1 + \#\{T_{\text{perm}} \geq T_{\text{observed}} - 10^{-14}\}}{200}, \qquad \text{reject if } p \leq 0.05. \tag{1}$$

The 199 label permutations and final batches are shared across methods. The kernel bandwidth is fixed using selection data. Figure 1 measures detection after selection; higher power is better. The two references distinguish selection errors from testing difficulty and noise dilution.

For mean differences, global power is 0.928, versus 0.992 for individual-coordinate MMD screening; the paired difference is -6.3 percentage points [-14.3, -1.3] ($n = 30$). All 30/30 mean-shift global searches meet the optimization tolerance, yet greedy power is 0.965. Solving the empirical robust objective accurately does not maximize this independent test's power.

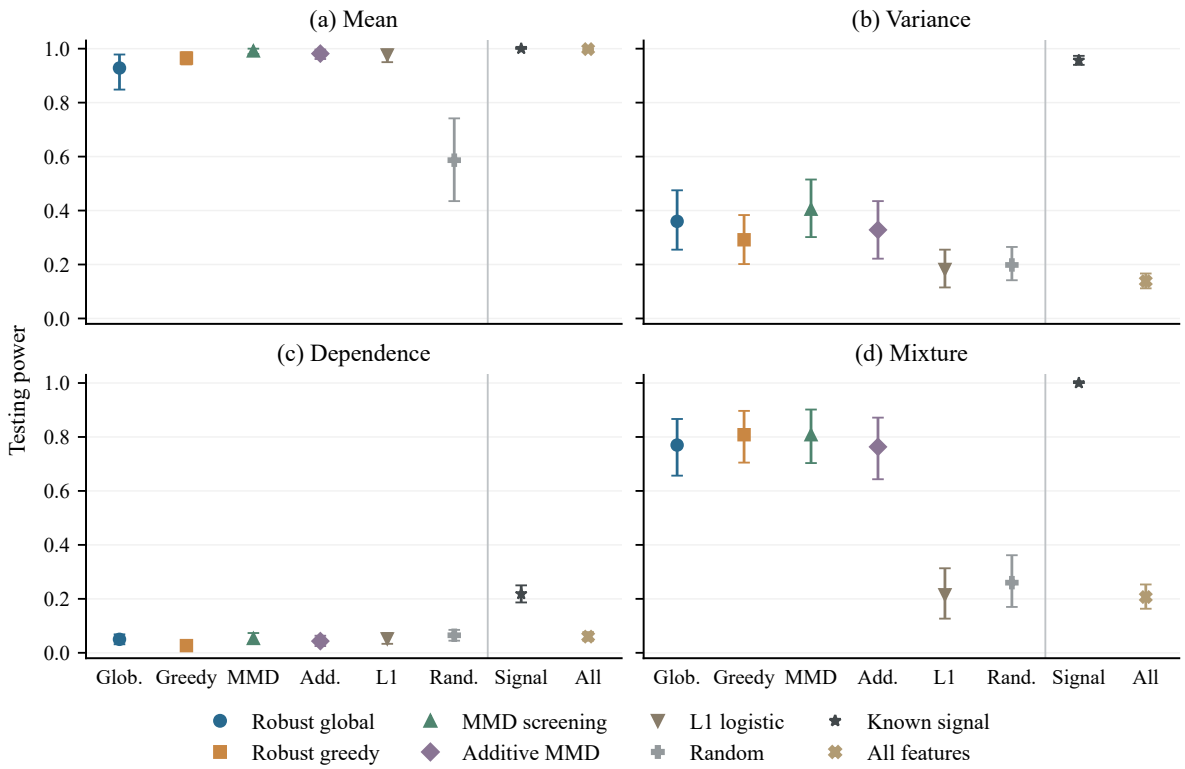



**Figure 1:** Common-test power (higher is better). The six selectors use three variables; Signal uses the true three and All uses twelve. Points and bars are means and 95% seed-bootstrap intervals.

The known-signal powers are 0.957 for variance and 1.000 for mixture differences, whereas using every variable gives 0.138 and 0.207. Keeping all the signal is not enough when irrelevant coordinates weaken the common test. In the dependence case, even the known-signal power is 0.218: the failure of practical selectors is accompanied by limited sensitivity of the 40-observation-per-class test.

Table 3 tests two samples from the same population. Their rejection rate measures false alarms, which should not systematically exceed 0.05; lower conservative rates are acceptable.

**Table 3:** False-alarm rate of the common RBF test (declared level 0.05). Entries average testing batches within each training seed, then the 30 seeds. Known signal and all features are diagnostic references.

| Selector | Mean | Variance | Dependence | Mixture |
|---|---|---|---|---|
| Global risk | 0.052 | 0.040 | 0.048 | 0.055 |
| Robust greedy | 0.053 | 0.047 | 0.060 | 0.055 |
| MMD screening | 0.050 | 0.048 | 0.052 | 0.052 |
| Additive MMD | 0.050 | 0.050 | 0.047 | 0.058 |
| L1 logistic | 0.043 | 0.045 | 0.050 | 0.050 |
| Random | 0.048 | 0.050 | 0.047 | 0.052 |
| Known signal | 0.053 | 0.050 | 0.055 | 0.048 |
| All features | 0.055 | 0.057 | 0.050 | 0.058 |

### 2.1 What changes when the learned weights are retained?

The common test uses joint similarity on the selected coordinates. Wang's additive method instead combines coordinate-wise statistics, choosing weights through

$$\max_{w} \; t^{\top} w - \lambda w^{\top} \hat{\Sigma} w, \qquad \|w\|_2 = 1, \quad \|w\|_0 \le d.$$

Here $t_j$ is the training MMD statistic for coordinate $j$, $\hat{\Sigma}$ estimates their joint variability, $w_j$ is its weight, and $d = 3$ limits nonzero weights. The first term rewards a measured difference; the second penalizes variability, with fixed $\lambda = 0.1$. Weights may be negative, so their similarity combination need not be a positive-semidefinite kernel. For the same selected coordinates, common power drops the weights, while native power retains the signed additive weights. The native permutation test uses the raw weighted MMD, without the training variance penalty. This is a partial implementation, not a complete reproduction.

**Table 4:** Two tests on the same Wang-selected support. Common RBF power uses only its coordinates; native power retains the signed additive weights. Entries are seed means and 95% intervals; native false alarms are also shown.

| Difference | $n$ | Common RBF power | Native power | Native false alarms |
|---|---|---|---|---|
| Mean | 30 | 0.982 [0.962, 0.997] | 0.977 [0.950, 0.998] | 0.045 [0.025, 0.065] |
| Variance | 30 | 0.328 [0.222, 0.435] | 0.518 [0.408, 0.625] | 0.052 [0.033, 0.072] |
| Dependence | 30 | 0.043 [0.025, 0.063] | 0.045 [0.030, 0.060] | 0.052 [0.035, 0.070] |
| Mixture | 30 | 0.763 [0.643, 0.872] | 0.945 [0.915, 0.970] | 0.040 [0.025, 0.057] |

The native test is substantially stronger in the variance and mixture cases. Treating common RBF power as the performance of the complete additive method would understate that method. The common comparison instead isolates the usefulness of the selected coordinates.

## 3 How many irrelevant variables are selected, and how many signals are missed?

A test can detect a difference without recovering every informative variable. We therefore also compare the chosen set $S$ with the true signal set $T_*$. The false discovery proportion (FDP) is the fraction of selected variables that are irrelevant. The non-discovery proportion (NDP) is the fraction of true signals that are missed. Both should be small:

$$\mathrm{FDP} = \frac{|S \setminus T_*|}{\max\{1, |S|\}}, \qquad \mathrm{NDP} = \frac{|T_* \setminus S|}{|T_*|}. \tag{2}$$

For the six practical selectors, $|S| = |T_*| = 3$, so FDP and NDP are equal on each run. A single plotted value reports both, rather than duplicating the same data. Uniform random selection has expected FDP and NDP 0.75; perfect recovery has value zero. The all-feature reference instead has FDP 0.75 and NDP zero, because it retains all three signals and all nine noise variables. Figure 2 now asks whether those useful coordinates are the true informative ones. Smaller FDP/NDP is better. Values near the random expectation 0.75 indicate little recovery beyond uniformly choosing three coordinates. For mean differences, global FDP/NDP is 0.467, versus 0.244 for MMD screening. For mixture differences, global FDP/NDP is 0.367, versus 0.289 for MMD screening. Thus a high rejection probability can coexist with selecting irrelevant variables: the testing and recovery questions are different.

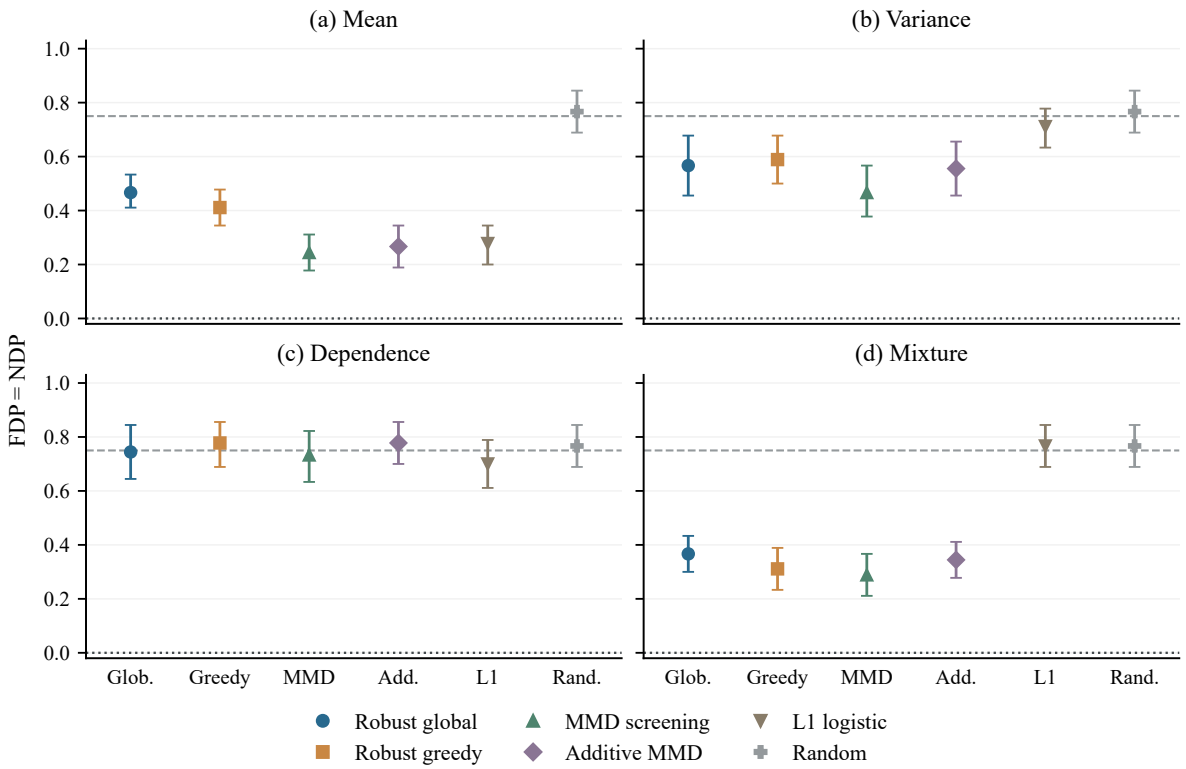



**Figure 2:** Variable-recovery errors: smaller is better. FDP and NDP coincide for the six three-variable selectors. The lines show perfect recovery (0) and the expected random error (0.75); the observed random selector is a separate point. Points and intervals use training seeds.

## 4 Do the selected variables support prediction on new observations?

We next hold the prediction model fixed and vary only its selected inputs. A Gaussian-kernel support-vector machine (SVM) is fitted on a separate sample for every selected subset. Its balanced error is the mean of the two classwise error rates; smaller is better, and 0.5 is chance performance. Figure 3 compares the same independently fitted SVM on the six practical supports. Lower balanced error means better prediction. This does not reuse the robust selector's fitted test. For mean differences, errors are 0.315 for global selection and 0.279 for MMD screening. For mixture differences, errors are 0.314 for global selection and 0.289 for MMD screening. The robust selection objective does not consistently produce the best inputs for this different prediction model.

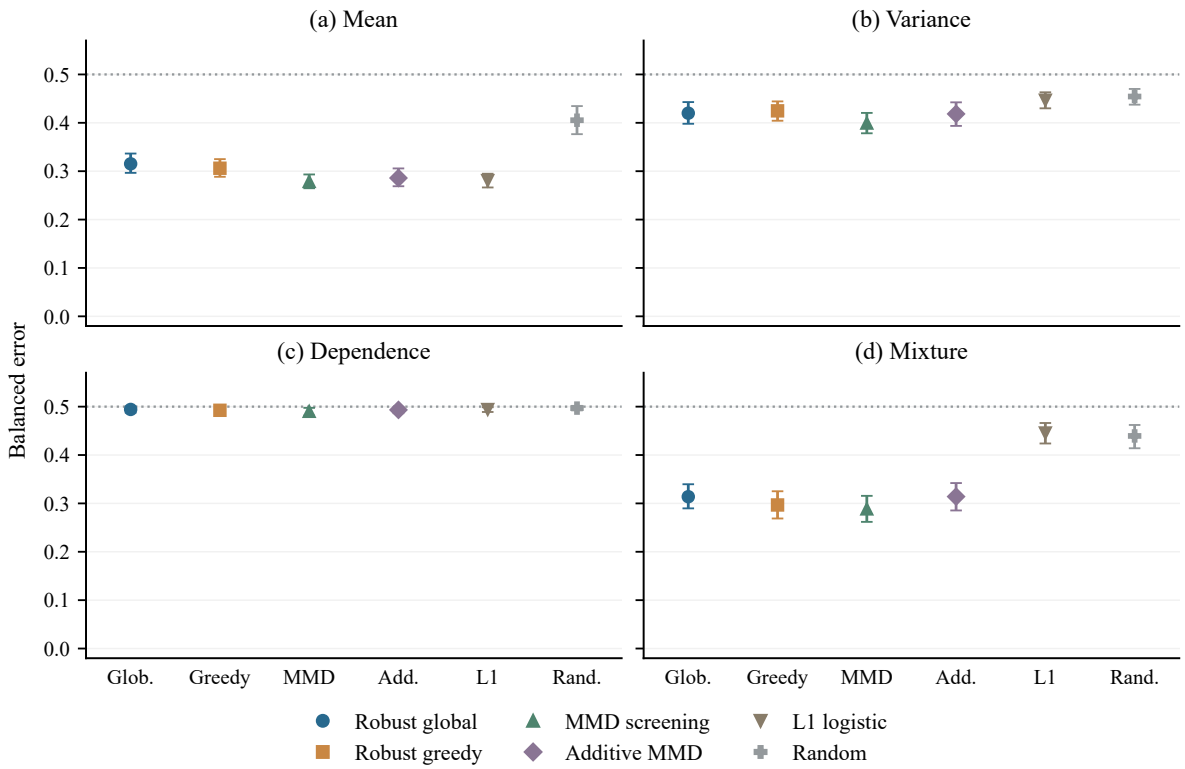



**Figure 3:** Independent SVM balanced error: smaller is better and 0.5 is chance performance. All six selectors use the same fitting samples and model settings. Points are seed means and bars are 95% intervals.

### 4.1 How the fitted robust bound compares with fresh-data error

A separate robust classifier is also refitted on each selected support. Let $p_0(x)$ be its probability of assigning class zero. A randomized decision draws its class from that probability. Its balanced error and the SVM's hard-decision balanced error are

$$\begin{aligned} \mathrm{BER}_{\mathrm{hard}} &= \tfrac{1}{2}(\mathrm{FPR} + \mathrm{FNR}), \\ \mathrm{BER}_{\mathrm{random}} &= \tfrac{1}{2}\{\mathbb{E}_0[1 - p_0(X)] + \mathbb{E}_1[p_0(Y)]\}. \end{aligned} \tag{3}$$

FPR and FNR are the two classwise wrong-decision rates. Unlike a hard classifier, a confident but non-deterministic probability still contributes randomized error. These two error definitions are not interchangeable. Table 5 compares the robust classifier's fitted bound with its randomized error on new data and under three shifts toward the other class. The shifts have magnitudes 0, 0.25 and 0.50; their maximum is a measured stress error, not a supremum over every distribution in a Wasserstein ball.

**Table 5:** Refitted robust classifier on globally selected variables. All errors are randomized balanced errors. The fitted bound is the classwise sum-risk upper bound divided by two. Stress is the within-seed maximum over three specified shifts; final columns count paired trials exceeding the fitted bound.

| Difference | $n$ | Fitted bound | Fresh error | Shift maximum | Above bound |
|---|---|---|---|---|---|
| Mean | 30 | 0.251 | 0.448 | 0.462 | 30/30 |
| Variance | 30 | 0.275 | 0.482 | 0.484 | 30/30 |
| Dependence | 30 | 0.287 | 0.497 | 0.501 | 30/30 |
| Mixture | 30 | 0.248 | 0.423 | 0.426 | 30/30 |

Fresh errors exceed the fitted empirical bound in 120/120 paired trials. This is not a certificate failure: the bound concerns the fitted empirical distribution ball, and the unknown population has not been shown

to belong to it. A low empirical worst-case loss cannot be read as a demonstrated population-error guarantee.

### 4.2 Changing the uncertainty radius

Table 6 changes the radius used both for selection and for refitting the robust classifier. It checks the whole procedure, rather than holding the selected variables fixed. The three radii are preset alternatives, not choices made from the final test outcomes.

**Table 6:** Joint radius sensitivity. Entries are seed means and 95% intervals; errors are randomized balanced errors. Stress is the maximum over preset shifts within each seed.

| Alternative | $\theta$ | $n$ | Power | Fresh error | Shift maximum |
|---|---|---|---|---|---|
| Mean | 0.10 | 10 | 0.790<br>[0.595, 0.915] | 0.494<br>[0.491, 0.497] | 0.496<br>[0.493, 0.499] |
| Mean | 0.25 | 10 | 0.855<br>[0.650, 0.980] | 0.466<br>[0.450, 0.480] | 0.474<br>[0.461, 0.485] |
| Mean | 0.50 | 10 | 0.955<br>[0.915, 0.990] | 0.426<br>[0.411, 0.440] | 0.442<br>[0.429, 0.453] |
| Mixture | 0.10 | 10 | 0.495<br>[0.295, 0.710] | 0.486<br>[0.478, 0.493] | 0.487<br>[0.480, 0.493] |
| Mixture | 0.25 | 10 | 0.630<br>[0.425, 0.845] | 0.446<br>[0.417, 0.469] | 0.448<br>[0.419, 0.471] |
| Mixture | 0.50 | 10 | 0.970<br>[0.930, 0.995] | 0.371<br>[0.352, 0.392] | 0.378<br>[0.357, 0.401] |

Larger radii improve the recorded power and randomized error in these two mechanisms. This is a result of the preset comparison, not a general monotonicity guarantee or an independently validated radius-selection rule.

## 5 What happens on real datasets?

WDBC distinguishes benign from malignant breast-cancer observations using 30 available variables; Ionosphere distinguishes good from bad radar returns using 34 variables [2, 3]. Each selector keeps three variables. The subsequent classifiers use disjoint fitting and final observations, as in the synthetic study. There is no known true signal set, so we report prediction rather than claim variable-recovery accuracy.

**Table 7:** Independent common classifiers on real datasets. Entries are mean ± overlapping-split SD. Robust error is randomized BER; SVM error is hard BER. These are incumbent supports when search is limited.

| Data | Selector | Robust randomized BER | SVM BER |
|---|---|---|---|
| WDBC | Global risk | 0.251±0.046 | 0.081±0.017 |
| WDBC | Robust greedy | 0.220±0.032 | 0.080±0.021 |
| WDBC | MMD screening | 0.179±0.013 | 0.084±0.017 |
| WDBC | Additive MMD | 0.180±0.013 | 0.082±0.014 |
| WDBC | L1 logistic | 0.230±0.034 | 0.078±0.020 |
| WDBC | Random | 0.310±0.066 | 0.156±0.078 |
| Ionosphere | Global risk | 0.300±0.045 | 0.251±0.094 |
| Ionosphere | Robust greedy | 0.285±0.035 | 0.202±0.081 |
| Ionosphere | MMD screening | 0.284±0.044 | 0.195±0.053 |
| Ionosphere | Additive MMD | 0.273±0.042 | 0.181±0.058 |
| Ionosphere | L1 logistic | 0.284±0.034 | 0.198±0.049 |
| Ionosphere | Random | 0.330±0.028 | 0.303±0.069 |

Global search is time-limited in 40/40 splits; its supports must not be called certified optima.

# 6 Does reducing full subset evaluations also reduce time?

The revised algorithm chooses a binary subset by solving a global integer master at every step, then fully solves its continuous robust-testing problem. It uses no prediction model or candidate pool. We compare tangent bounds with tangent and fraction bounds, counting full subset evaluations and total seconds. Time includes both continuous solves and the integer master. We rerun 16 existing configurations on 4 generated datasets, with 3 timings per method. Both current variants solve the global master every step; only retaining the fraction bounds differs. Counts and seconds sum within-configuration medians. This reused-setting ablation is not independent confirmation.

**Table 8:** Subset-search ablation (3 timings per configuration). The first two rows compare bounds within the current algorithm; the last two compare earlier complete search procedures.

| Search | Evaluations | Master (s) | Total (s) | Optimal runs |
|---|---|---|---|---|
| Tangent | 601 | 12.090 | 15.672 | 48/48 |
| Tangent + fraction | 446 | 15.701 | 18.509 | 48/48 |
| Earlier prediction + tangent | 598 | 7.558 | 11.627 | 48/48 |
| Earlier prediction + fraction | 453 | 8.547 | 11.779 | 48/48 |

Retaining fractions changes full evaluations by -25.8% and total time by +18.1%. The earlier procedures also differ in candidate pools and periodic master updates, so their comparison cannot isolate prediction alone. Figure 4 displays the current paired comparison. The extra integer-master cost offsets the saved inner work. The earlier complete procedures are faster in this cohort. Removing prediction simplifies the algorithm; these timings do not show that prediction has no value.

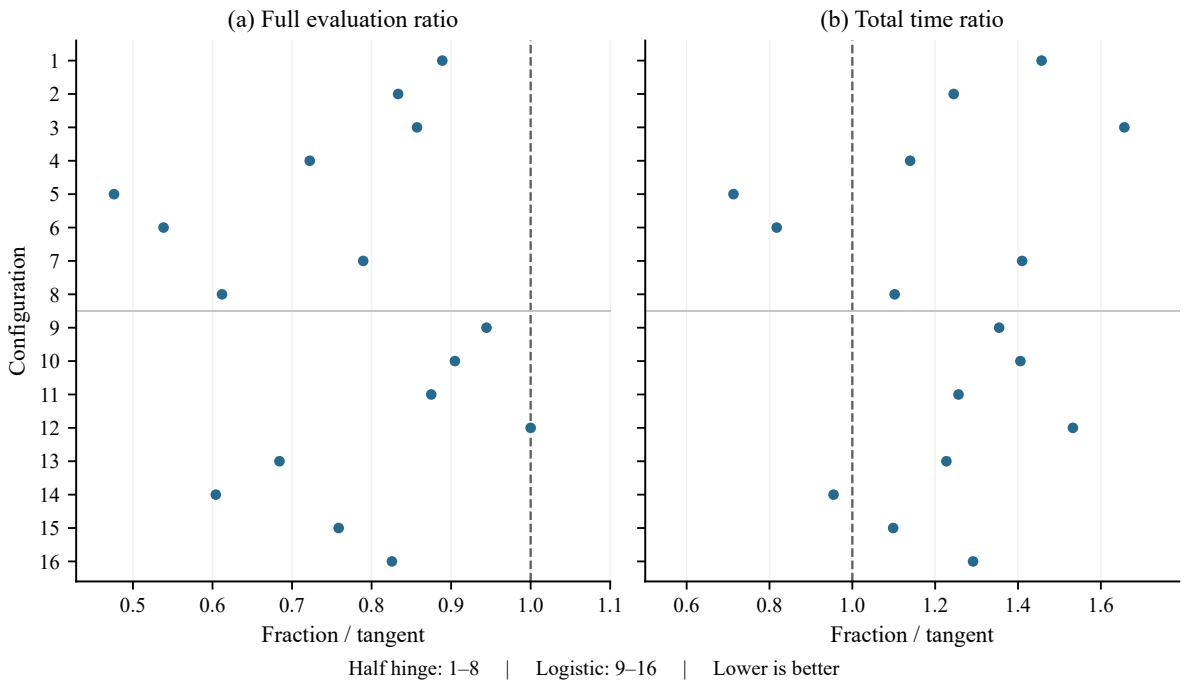



**Figure 4:** Fraction/tangent ratios for full evaluations and total time. Below one is better; one means equal cost. Each point is a within-configuration median ratio.

192/192 runs meet the numerical target; the maximum feasible-risk excess over the exhaustive lower reference is $1.6 \times 10^{-7}$. These small cases establish neither large-instance speed nor new statistical performance. Earlier failures and prediction comparisons remain in `docs/experiment_development_record.md`.

## 6.1 Earlier comparison as the dimension increases

The earlier implementation also compared three price constraints used to link the binary subset with the continuous problem: individual coordinate caps, a cap on their weighted sum, and a cap scaled by the risk being minimized. The last is tested with and without its fractional starting stage. All four versions solve the same binary robust objective. Across 159 solved gate/enumeration pairs, the largest recorded feasible-objective difference is 0.000, with 0 incompatible intervals at the combined solver tolerances. At $D \geq 20$, global risk reaches the limit in 15/15 mean instances. At $D \geq 20$, global risk reaches the limit in 15/15 mixture instances.

**Table 9:** $D = 12$ reference comparison. Time is charged seconds with 95% seed-bootstrap intervals; supports are distinct integer subsets, excluding fractional root calls.

| Alternative | Search | Solved | Time (s) | Supports |
|---|---|---|---|---|
| Mean | Box | 10/10 | 4.07 [3.57, 4.88] | 220.0 |
| Mean | Triangle | 10/10 | 4.63 [4.16, 5.08] | 219.9 |
| Mean | Risk, no root | 10/10 | 13.73 [13.05, 14.32] | 203.2 |
| Mean | Global risk | 10/10 | 16.54 [15.74, 17.28] | 197.8 |
| Mixture | Box | 10/10 | 3.52 [3.40, 3.63] | 220.0 |
| Mixture | Triangle | 10/10 | 4.75 [4.38, 5.08] | 219.2 |
| Mixture | Risk, no root | 10/10 | 13.22 [12.30, 14.07] | 200.0 |
| Mixture | Global risk | 10/10 | 16.39 [15.24, 17.52] | 197.0 |

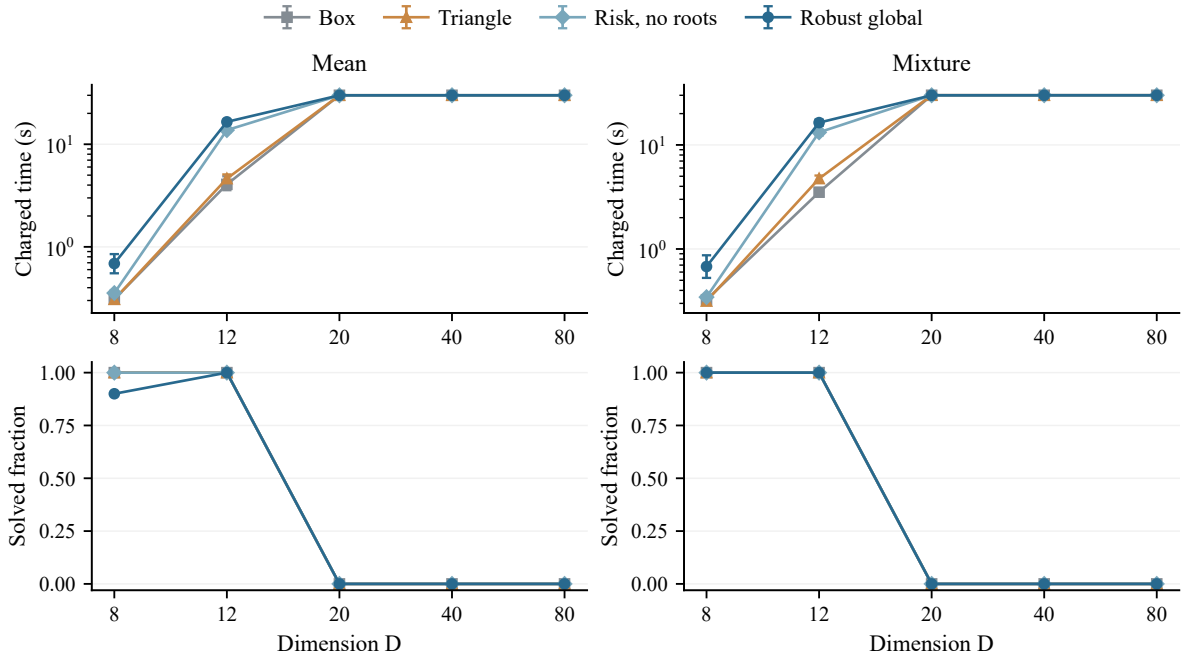



**Figure 5:** Earlier implementation: charged time and solved fraction as dimension grows. Lower time and higher solved fraction are preferable; unsuccessful runs remain included.

## 7 Implementation parameters and complete records

The following settings identify the experiments above. They are fixed before the final observations are evaluated. The main synthetic comparison uses 30 training seeds; each radius comparison uses 10. Synthetic intervals are 95% percentile bootstrap intervals over the seeds of the corresponding experiment, using 2,000 resamples after averaging test batches within each seed. Real-data entries use mean $\pm$ split standard deviation because the splits overlap. All time limits and failures remain in the corresponding denominators and charged computation.

**Table 10:** Reference statistical data. Selection and classifier fitting counts are per class; final counts show their unit. Repeats are training seeds or overlapping real-data splits.

| Data | $D/d$ | Selection / fitting per class | Final observations | Repeats |
|---|---|---|---|---|
| Synthetic | 12/3 | 20/20 | 1000/class | 30 |
| WDBC | 30/3 | 20/20 | 489 total | 20 |
| Ionosphere | 34/3 | 20/20 | 271 total | 20 |

**Table 11:** Reference optimization settings. Each row is evaluated separately for mean and mixture alternatives.

| $D$ | $d$ | Atoms/class | Seeds | Alternatives |
|---|---|---|---|---|
| 8 | 2 | 8 | 10 | Mean, mixture |
| 12 | 3 | 8 | 10 | Mean, mixture |
| 20 | 3 | 10 | 5 | Mean, mixture |
| 40 | 3 | 10 | 5 | Mean, mixture |
| 80 | 3 | 10 | 5 | Mean, mixture |

The primary class radii are both 0.25. Each synthetic seed has 20 alternative and 20 null batches, with 40 observations per class in a batch. Radius sensitivity uses 10 seeds per mechanism and radius.

Centering and sample-standard-deviation scaling use pooled selection observations and are then fixed. The common RBF uses $b^2$ equal to their median squared distance. SVMs use $C = 1$ and a median bandwidth from their own fitting sample. The robust classifier uses the selection loss and radius.

Additive MMD uses fixed $\lambda = 0.1$, without cross-validation. Supports are enumerated at $D \leq 12$; larger cases use forward selection and at most ten improving swaps. Testing bandwidths never use final labels. L1 logistic ranking uses liblinear, $C = 1$, and at most 2,000 iterations. At its time limit, greedy keeps its last completed stage and fills any remaining slots by MMD screening.

The global and greedy robust selectors each have a 30-s selection allowance. The robust radius is 0.25; numerical gap tolerance $10^{-6} + 10^{-5}|\mathrm{UB}|$ is used only by global search. The remaining selectors have no enforced wall-clock cutoff. The earlier global search starts from MMD screening and uses five fractional starting rounds. Randomized robust predictions use their fitted class probabilities; SVM hard decisions use threshold zero, with ties assigned to class zero.

**Table 12:** Fixed settings of the reused-configuration search ablation.

| Item | Setting |
|---|---|
| Cases and data | 16 configurations on 4 datasets; $(D, d) = (7, 2), (10, 3)$; 6 observations/class; seeds 19,23. |
| Loss and radius | Half hinge or logistic; radii $(0.12, 0.20)$ or $(0.30, 0.45)$. |
| Generated coordinates | Gaussian class zero; two class-one means shifted by 1.4,0.9; nuisance scales 0.3–3; no fitted standardization. |
| Common starting subset | $\lvert\bar{x}_{0j} - \bar{x}_{1j}\rvert/(s_{0j} + s_{1j} + 10^{-12})$; sample SD with ddof=0, index ties; no true signal identities. |
| Time and precision | 60 s, one thread; $\varepsilon = 2 \times 10^{-5}$, relative tolerance zero; inner/master targets $\varepsilon/4$. |
| Current subset choice | Global integer master at every step; tangent bounds, with or without stored fractions; no prediction, pool, or fractional root calls. |

Counts use distinct fully evaluated binary subsets; each current evaluation targets $\varepsilon/4$. Physical refinements and master time remain in the raw traces. Three timings per configuration rotate execution order. Earlier prediction settings remain documented with their source snapshots.

Reference computations use AMD Ryzen 9 7945HX with Radeon Graphics, one solver thread and SciPy-backed HiGHS 1.12.0; logistic checks use CVXPY 1.9.3 and Clarabel 0.11.1.

The earlier study's 1640 records include 3 failures and 189 time limits; complete metrics, supports and traces remain in `results/research_20261003`. Historical comparisons and failures are indexed in `docs/experiment_development_record.md`. The new ablation is separate under `reports/verification/20261012/no_prediction`; manifests identify all included inputs.

## References


[1] J. Wang, S. S. Dey, and Y. Xie. *Variable Selection for Kernel Two-Sample Tests*. 2024. arXiv:2302.07415v4. Code inspected at commit `031f740e9ca6` of MMDVar_Selection.

[2] UCI Machine Learning Repository. *Breast Cancer Wisconsin (Diagnostic)*. DOI:10.24432/C5DW2B.

[3] UCI Machine Learning Repository. *Ionosphere*. DOI:10.24432/C5W01B.